

V FOODS — Complete Campus Dining & Pre-Order System Guide

A plain-English operational and architectural guide to the V FOODS platform.

Designed for students, canteen franchisees, university dining administrators, and technical partners.

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1. What is V FOODS?

V FOODS is a smart food pre-ordering, prepaid digital wallet, and kitchen management platform custom-built for collegiate campuses and large festival dining grids.

The Problem on Campus

During 15-minute lecture breaks and lunch rush hours: - Hundreds of students crowd physical canteen counters simultaneously. - Payment terminals fail due to mobile network congestion or UPI timeouts. - Counter lines move slowly while cashiers manually handle payments, verify slips, and call out orders. - Students spend 25–40 minutes waiting in line, often missing meals or arriving late to classes.

The V FOODS Solution

1. **Pre-Order Ahead:** Students browse menus, customize items, and place pre-orders before walking to the food court.
2. **Prepaid Digital Wallet:** Payments are deducted in **under 20 milliseconds** from a pre-loaded campus wallet, eliminating checkout failure rates.
3. **Kitchen Display System (KDS):** Kitchen staff receive orders digitally on a live board, prepare meals in advance, and package them hot.

4. **Instant Token Counter Collection:** Students receive a **4-digit digital token** (e.g. #4826) and QR pass. They arrive at the counter, show their token, pick up their hot meal in seconds, and leave. **Zero waiting in lines.**
-

2. Full-Mode System Architecture

The following diagram illustrates how all components of V FOODS communicate in real time across the university network:

flowchart TB

```
subgraph Clients["📱 CLIENT LAYER (Progressive Web App)"]
    direction TB
    C1["👤 Student App<br/>(Browse, Wallet, Pre-Order)"]
    C2["👨🍳 Kitchen Staff KDS<br/>(Live Ticket Dispatch)"]
    C3["👤 Shop Owner Dashboard<br/>(Sales, Menu, Staff)"]
    C4["👑 Super Admin Mission Control<br/>(Campus Oversight & Audit)"]
end

subgraph CDN["⚡ HOSTING & EDGE (Render Global CDN)"]
    direction TB
    Vite["Vite + React 19 Frontend<br/>(Cached globally, Offline Service Worker)"]
end

subgraph API["🚀 APPLICATION BACKEND (FastAPI / Python 3.11)"]
    direction TB
    F1["FastAPI Core Engine"]
    F2["PhonePe Webhook Verifier"]
    F3["HMAC-SHA256 Token QR Engine"]
    F4["Background Task Scheduler"]
end

subgraph DataLayer["🗄️ DATA & REALTIME LAYER (Supabase Cloud)"]
    direction TB
    Postgres["PostgreSQL 15 Database"]
    RLS["Row-Level Security (RLS)"]
    Realtime["Realtime WebSockets (Live Sync)"]
    RPC["Atomic Stored Procedures<br/>(credit_wallet, place_order_wallet)"]
end

subgraph Gateways["🌐 THIRD-PARTY INTEGRATIONS"]
    PhonePe["👛 PhonePe PG Gateway<br/>(UPI, NetBanking, Cards)"]
    MSG91["📠 MSG91 Notification Engine<br/>(WhatsApp & SMS Alerts)"]
end

%% Connections
Clients -->|HTTPS Requests| Vite
Vite -->|Client State| API
Vite -->|Direct Read/Write| Postgres
API -->|Service Role DB Access| Postgres
Postgres --- RLS
Postgres --- RPC
Postgres -->|Live Database Changes| Realtime
Realtime -. ->|Instant UI Updates| Clients
API -->|Initiate Payment| PhonePe
PhonePe -->|Webhook Verification| API
API -->|Order Ready / OTP| MSG91
MSG91 -. ->|SMS / WhatsApp| C1
```

3. The End-to-End Order Lifecycle

Every pre-order follows a structured 4-step pipeline designed for zero physical friction:

```
sequenceDiagram
    autonumber
    actor Student as 🧑 Student
    participant App as 📱 V FOODS App
    participant Wallet as 💰 Campus Wallet
    participant DB as 📄 Database (Supabase)
    participant KDS as 🍳 Kitchen KDS
    actor Staff as 👤 Kitchen Staff

    Student->>App: Browse Food Court & Select Dishes
    Student->>App: Choose Timing (⚡ As soon as ready OR 🕒 Slot)
    App->>Wallet: Verify Sufficient Balance
    Wallet->>DB: Atomic RPC: Deduct Wallet & Create Order
    DB-->>App: Order Confirmed + 4-Digit Token (#4826)
    DB->>KDS: Realtime WebSocket: New Order Event (Audio Chime)
    Staff->>KDS: Tap 'Start Cooking' → Status: PREPARING
    Staff->>KDS: Pack Food & Tap 'Mark Ready' → Status: READY
    DB-->>App: Push & SMS Notification: 'Your Food is Ready at Counter!'
    Student->>Staff: Show 4-Digit Token / QR Pass at Counter
    Staff->>KDS: Verify & Tap 'Mark Collected'
    DB-->>App: Order Completed & Archived
```

4. Dual Payment Architecture: Wallet + Instant Gateway

V FOODS provides flexible dual payment channels right at the checkout counter:

```
flowchart TD
    Cart["Student Cart Assembly"] --> Choice{"Select Payment Option"}

    %% Option 1: Campus Wallet
    Choice -->|Option 1: Prepaid Wallet| Wallet["V FOODS Campus Wallet"]
    Wallet -->|If Balance Sufficient| WPay["1-Tap Atomic Wallet Debit (< 20ms)"]
    Wallet -->|If Low Balance| WTop["1-Tap Top-Up Deficit via PhonePe / UPI"]
    WTop --> WPay
    WPay --> OrderSuccess["Order Confirmed + Token (#4826) → Kitchen KDS"]

    %% Option 2: Instant Gateway
    Choice -->|Option 2: Instant Gateway| Gateway["Instant Payment Gateway"]
    Gateway --> UPIOptions["PhonePe UPI / Paytm UPI / GPay / Cards"]
    UPIOptions --> BankAuth["Direct Bank Authorization via Gateway"]
    BankAuth --> OrderSuccess
```

Payment Option Comparison:

Feature	Option A: V FOODS Campus Wallet	Option B: Instant Payment Gateway
Best For	Daily meals, regular students, 10-min rush breaks	Visitors, first-time students, direct bank payers
Speed	Under 20 milliseconds (Fastest)	5–15 seconds (Standard UPI app authorization)
Prerequisites	Pre-loaded wallet balance (rechargeable anytime)	None; works with zero prior balance
Providers	Internal wallet float (recharged via PhonePe/UPI)	PhonePe, Paytm, Google Pay, BHIM, Cards
Network Reliability	Works seamlessly even with spotty campus signals	Requires active mobile internet to authorize UPI
Deficit Support	Auto-calculated deficit top-up (+ ₹X & Pay)	Not needed — charges exact order total directly

5. The Four Specialized Dashboards

V FOODS provides 4 distinct interfaces tailored to specific responsibilities:

```
graph TD
  subgraph Roles["CAMPUS USER ACCESS LEVELS"]
    direction TB
    R1["1. Student / User<br/>• Mobile PWA Layout<br/>• 3-Column Food Grid<br/>• Left Rail Food"]
    R2["2. Kitchen Staff<br/>• Live KDS Kanban Board<br/>• Placed → Preparing → Ready<br/>• Co"]
    R3["3. Canteen Owner<br/>• Daily Gross & Net Revenue<br/>• Menu Catalog & Pricing Editor<br/"]
    R4["4. Super Admin<br/>• Campus-wide GMV & 5% Fee Ledger<br/>• 13 Canteen Health Matrix<br/>"]
  end
```

6. Campus Canteens & Food Courts

V FOODS models the physical layout of **VIT Chennai** into **5 Major Food Courts** housing **13 distinct counters**:

```

graph LR
    subgraph Campus["VIT CHENNAI CAMPUS"]
        FC1["Gazebo Food Court<br/>(Central Campus Plaza)"]
        FC2["North Square Food Court<br/>(North Square Building)"]
        FC3["AB3 Food Complex<br/>(Academic Block 3 Courtyard)"]
        FC4["Academic Blocks Diner<br/>(Delta & AB1 Walkways)"]
        FC5["Campus Pavilion & Stalls<br/>(Lakeview Activity Center)"]
    end

    FC1 --> G1["Gazebo C1 – Snacks & Fast Food"]
    FC1 --> G2["Gazebo C2 – Desserts & Sweets"]
    FC1 --> G3["Dakshin Chitra – South Meals"]
    FC1 --> G4["Lassi House – Shakes & Beverages"]

    FC2 --> N1["Georgia Canteen"]
    FC2 --> N2["Alpha Non-Veg Kitchen"]
    FC2 --> N3["Sri's Fast Food"]
    FC2 --> N4["Juice & Rice Corner"]

    FC3 --> AB3["AB3 Amphitheatre Kitchen"]
    FC4 --> AB1["AB1 Walkway Canteen"]
    FC4 --> DELTA["Delta Snacks Counter"]
    FC5 --> AV["Aavin Milk & Ice Cream Booth"]
    FC5 --> VM["V Mart Provisional Store"]

```

7. Database Security & Row-Level Isolation (RLS)

All permissions in V FOODS are secured directly by **PostgreSQL Row-Level Security (RLS)** in Supabase. The database rejects unauthorized queries at the engine level:

SUPABASE POSTGRESQL ENGINE	
STUDENTS	→ Read only own profile, wallet transactions, orders
KITCHEN STAFF	→ Read and update orders belonging strictly to OUTLET_ID
CANTEEN OWNER	→ Edit menu, view revenue strictly for their OUTLET_ID
SUPER ADMIN	→ Full platform visibility and governance

Even if someone tampers with frontend code or runs unauthorized network commands, the database automatically filters out and blocks any record outside the user's validated session token.

8. Production Deployment & Infrastructure

V FOODS is deployed using automated continuous deployment on **Render.com**:

- **Frontend** (`vfoods-web`):

- React 19 + Vite PWA static site deployed globally via Render's CDN.
- Automatically rebuilds on git pushes to `main`.
- **Backend API (`vfoods-api`):**
- Python 3.11 FastAPI service deployed in the Singapore region (`uvicorn main:app`).
- Connects to Supabase PostgreSQL via connection pooling and verifies PhonePe webhook signatures.
- **Database (`Supabase Cloud`):**
- Hosted PostgreSQL 15 instance with automatic failover, WAL replication, and real-time subscription hubs.